
ASSIGNMENT REPORT

Course	Amazon Web Services (AWS)
Instructor	Nitha
Task ID	Task #48780
Assignment Title	Jenkins Basics
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Objectives

This assignment covers three core learning goals:

- Learn the basics of the Jenkins Automation Server and understand its core concepts.
- Learn about the CI/CD deployment methods commonly used with Jenkins.
- Install Jenkins on an AWS EC2 instance running RHEL 10 and verify the setup.

Part 1: Installing Jenkins on AWS EC2 (RHEL 10)

Create an instance and ssh into it

- An AWS EC2 instance running RHEL 10 (t2.micro).
- Security Group must allow inbound TCP on port 8080 (Jenkins UI) and port 22 (SSH).

Connect to the instance over SSH using your key pair:

```
ssh -i your-key.pem ec2-user@<PUBLIC_IP>
```

Update the system:

```
sudo dnf update -y
```

Check Available Java Versions

search to see what versions are available in the repository:

```
sudo dnf search openjdk
```

```
[ec2-user@ip-172-31-10-6 ~]$ sudo dnf update -y
Updating Subscription Management repositories.
Last metadata expiration check: 0:55:12 ago on Wed Apr 22 05:46:01 2026.
Dependencies resolved.
Nothing to do.
Complete!
[ec2-user@ip-172-31-10-6 ~]$ sudo dnf search openjdk
Updating Subscription Management repositories.
Last metadata expiration check: 0:55:18 ago on Wed Apr 22 05:46:01 2026.
===== Name & Summary Matched: openjdk =====
java-21-openjdk.x86_64 : OpenJDK 21 Runtime Environment
java-21-openjdk-demo.x86_64 : OpenJDK 21 Demos
java-21-openjdk-devel.x86_64 : OpenJDK 21 Development Environment
java-21-openjdk-headless.x86_64 : OpenJDK 21 Headless Runtime Environment
java-21-openjdk-javadoc.x86_64 : OpenJDK 21 API documentation
java-21-openjdk-javadoc-zip.x86_64 : OpenJDK 21 API documentation compressed in a single archive
java-21-openjdk-jmods.x86_64 : JMods for OpenJDK 21
java-21-openjdk-src.x86_64 : OpenJDK 21 Source Bundle
java-21-openjdk-static-libs.x86_64 : OpenJDK 21 libraries for static linking
java-25-openjdk.x86_64 : OpenJDK 25 Runtime Environment
java-25-openjdk-crypto-adapter.x86_64 : OpenJDK 25 Cryptography Adapter Library
java-25-openjdk-demo.x86_64 : OpenJDK 25 Demos
java-25-openjdk-devel.x86_64 : OpenJDK 25 Development Environment
java-25-openjdk-headless.x86_64 : OpenJDK 25 Headless Runtime Environment
java-25-openjdk-javadoc.x86_64 : OpenJDK 25 API documentation
java-25-openjdk-javadoc-zip.x86_64 : OpenJDK 25 API documentation compressed in a single archive
java-25-openjdk-jmods.x86_64 : JMods for OpenJDK 25
java-25-openjdk-src.x86_64 : OpenJDK 25 Source Bundle
java-25-openjdk-static-libs.x86_64 : OpenJDK 25 libraries for static linking
maven-openjdk21.noarch : maven binding for openjdk21
maven-openjdk25.noarch : maven binding for openjdk25
```

dnf update completes successfully; dnf search openjdk lists available Java versions

Install Java 25 (OpenJDK)

From the search results, Java 25 (LTS) is the recommended version for Jenkins 2.55. Install it with:

```
sudo dnf install -y java-25-openjdk.x86_64
```

Verify the installation:

```
java -version
```

Add the Jenkins YUM/DNF Repository

Jenkins is not in the default RHEL repositories. You must add the official Jenkins repository. Download the .repo file:

```
sudo wget -O /etc/yum.repos.d/jenkins.repo \
https://pkg.jenkins.io/redhat-stable/jenkins.repo
```

The wget command fetches the repository configuration file and saves it to /etc/yum.repos.d/jenkins.repo. You will see a 301 redirect followed by a 200 OK and a download progress bar.

Import the Jenkins GPG Signing Key

Import the Jenkins GPG key to authenticate packages from the Jenkins repository:

```
sudo rpm --import https://pkg.jenkins.io/redhat-stable/jenkins.io-2023.key
```

```
[ec2-user@ip-172-31-10-6 ~]$ java --version
openjdk 25.0.2 2026-01-20 LTS
OpenJDK Runtime Environment (Red_Hat-25.0.2.0.10-3) (build 25.0.2+10-LTS)
OpenJDK 64-Bit Server VM (Red_Hat-25.0.2.0.10-3) (build 25.0.2+10-LTS, mixed mode, sharing)
```

```
[ec2-user@ip-172-31-10-6 ~]$ sudo wget -O /etc/yum.repos.d/jenkins.repo \
https://pkg.jenkins.io/redhat-stable/jenkins.repo
--2026-04-22 06:42:24-- https://pkg.jenkins.io/redhat-stable/jenkins.repo
Resolving pkg.jenkins.io (pkg.jenkins.io)... 151.101.210.133, 2a04:4e42:31::645
Connecting to pkg.jenkins.io (pkg.jenkins.io)|151.101.210.133|:443... connected.
HTTP request sent, awaiting response... 301 Moved Permanently. [6] Install Jenkins
Location: https://pkg.jenkins.io/rpm-stable/jenkins.repo [following]
--2026-04-22 06:42:24-- https://pkg.jenkins.io/rpm-stable/jenkins.repo
Reusing existing connection to pkg.jenkins.io:443.
HTTP request sent, awaiting response... 200 OK
Length: 267 [application/octet-stream]
Saving to: '/etc/yum.repos.d/jenkins.repo'

/etc/yum.repos.d/jenkins.repo 100%[=====] 267 --.-KB/s in 0s

2026-04-22 06:42:24 (9.87 MB/s) - '/etc/yum.repos.d/jenkins.repo' saved [267/267]

[ec2-user@ip-172-31-10-6 ~]$ ls
[ec2-user@ip-172-31-10-6 ~]$ sudo rpm --import https://pkg.jenkins.io/redhat-stable/jenkins.io-2023.key
warning: Certificate SBA31D57EF5975CA:
  The certificate is expired: The primary key is not live
  Subkey SBA31D57EF5975CA is expired: The primary key is not live
  Certificate does not have any usable signing keys
```

Java 25 already installed; Jenkins repo file downloaded successfully to /etc/yum.repos.d/jenkins.repo; GPG key import shows a certificate warning (key expired) but proceeds normally

Install Jenkins

With the repository and key in place, install Jenkins using dnf:

```
sudo dnf install jenkins -y
```

If Jenkins is already installed, dnf will report 'Package jenkins-2.555.1-1.noarch is already installed. Nothing to do. Complete!'

Enable and Start Jenkins Service

Enable Jenkins to start automatically on boot and start it immediately:

```
sudo systemctl enable --now jenkins
```

This single command both enables the service (start on boot) and starts it right now. You can verify the status with:

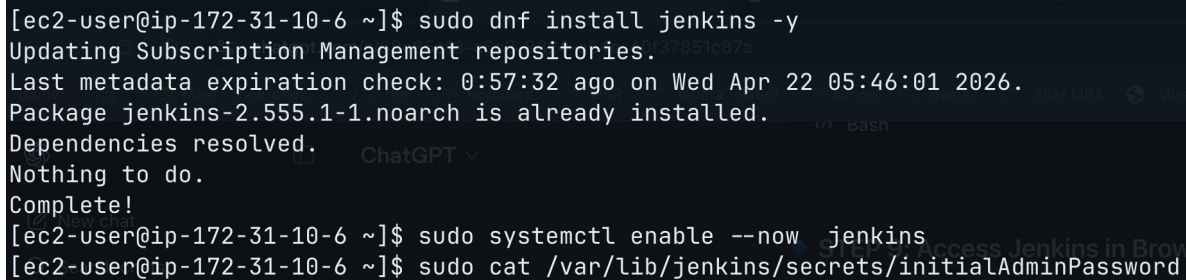
```
sudo systemctl status jenkins
```

Step 9: Retrieve the Initial Admin Password

On first launch, Jenkins generates a one-time Administrator password stored in a file. Retrieve it with:

```
sudo cat /var/lib/jenkins/secrets/initialAdminPassword
```

Copy the alphanumeric string that is printed. You will need it in the next step to unlock Jenkins.

A terminal window screenshot showing the installation of Jenkins on an EC2 instance. The user runs 'sudo dnf install jenkins -y', which updates repositories and confirms the package is already installed. Then, the user runs 'sudo systemctl enable --now jenkins'. Finally, the user runs 'sudo cat /var/lib/jenkins/secrets/initialAdminPassword', which outputs a long alphanumeric string: 'J37851c07a'.

```
[ec2-user@ip-172-31-10-6 ~]$ sudo dnf install jenkins -y
Updating Subscription Management repositories.
Last metadata expiration check: 0:57:32 ago on Wed Apr 22 05:46:01 2026.
Package jenkins-2.555.1-1.noarch is already installed.
Dependencies resolved.
Nothing to do.
Complete!
[ec2-user@ip-172-31-10-6 ~]$ sudo systemctl enable --now jenkins
[ec2-user@ip-172-31-10-6 ~]$ sudo cat /var/lib/jenkins/secrets/initialAdminPassword
J37851c07a
```

cat retrieves the initialAdminPassword

Open Port 8080 in EC2 Security Group

In the AWS console, navigate to EC2 > Security Groups > your instance's security group. Add an inbound rule:

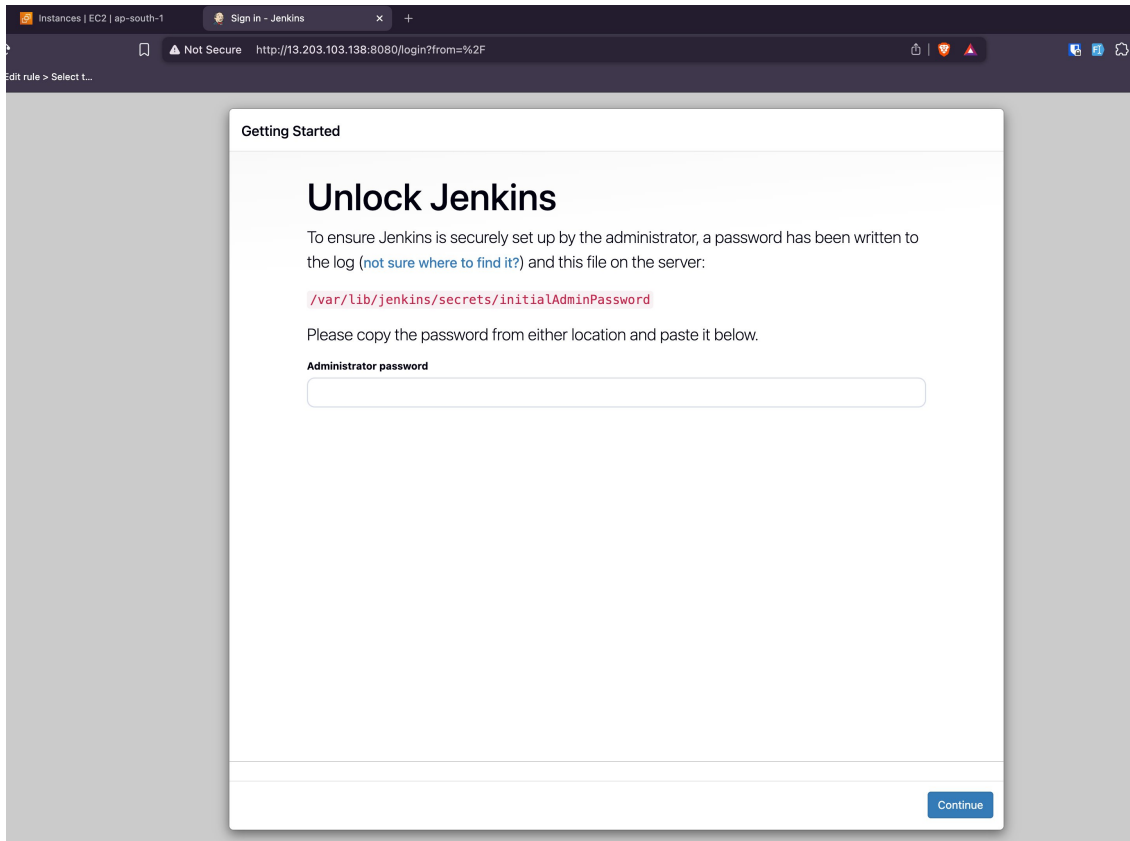
- Type: Custom TCP
- Port: 8080
- Source: 0.0.0.0/0 (or your IP for security)
- Description: Jenkins-port

Access Jenkins in the Browser

Open a web browser and navigate to:

```
http://<EC2_PUBLIC_IP>:8080
```

Replace <EC2_PUBLIC_IP> with the public IP address of your EC2 instance (visible in the AWS console, e.g., 13.203.103.138).

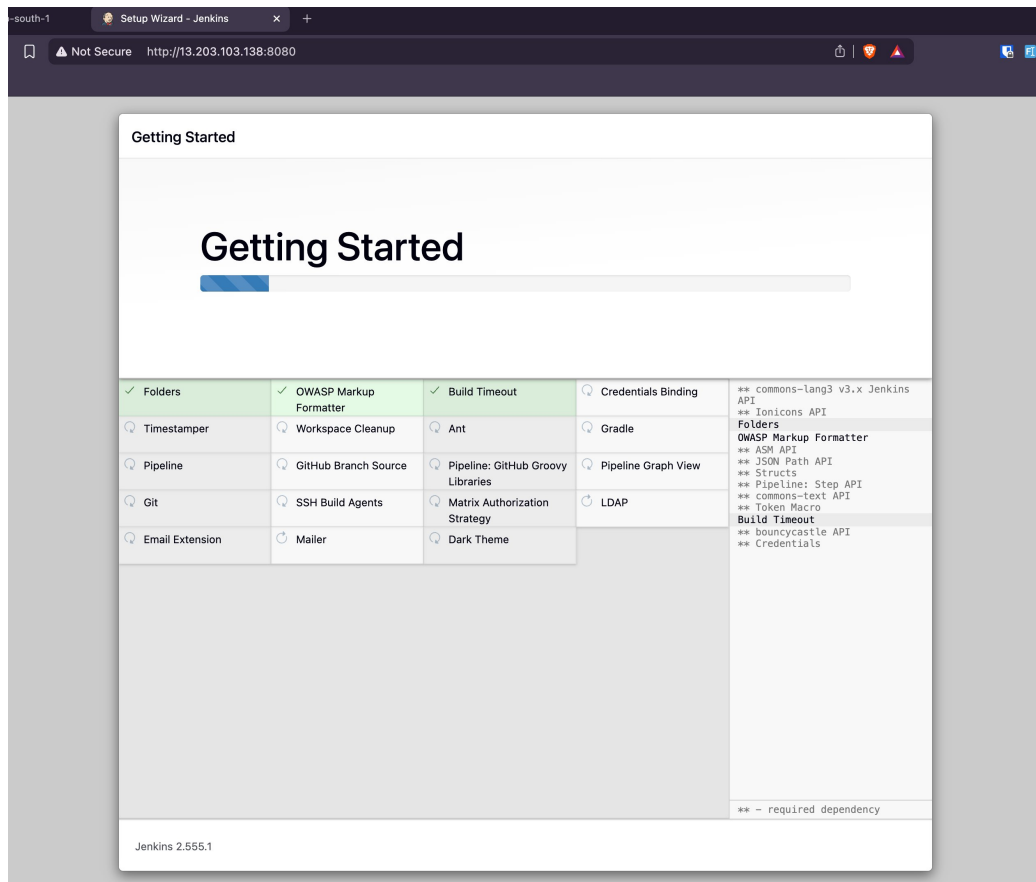


Jenkins 'Unlock Jenkins' screen at <http://13.203.103.138:8080>. Paste the `initialAdminPassword` copied and click *Continue*.

Complete the Setup Wizard

After entering the initial admin password:

1. **Install Suggested Plugins** – Click 'Install suggested plugins'. Jenkins will automatically install the most commonly used plugins (Git, Pipeline, Credentials Binding, Matrix Authorization Strategy, etc.). A progress screen shows each plugin being installed.
2. **Create Admin User** – After plugin installation, set up your permanent admin username, password, and email address.
3. **Configure Jenkins URL** – Confirm the Jenkins URL (the public IP:8080 address) and click 'Save and Finish'.
4. **Start Using Jenkins** – Click 'Start using Jenkins' to enter the Jenkins dashboard.



Jenkins Setup Wizard installing suggested plugins, Jenkins version 2.555.1 shown at the bottom.